

Analyzing the Impact of Three-Point Efficiency on NBA Team Performance

Aansh Jha

University of Connecticut

May 12, 2025

Abstract

Over the past four decades, the three-point shot has evolved from a niche option into a fundamental component of NBA offense. This study provides a comprehensive, data-driven assessment of how three-point shooting volume and efficiency influence team success. Using team-level statistics from the 1980-2024 seasons (covering the entire three-point era), I documented long-term trends and used regression models to quantify the impact of three-point attempts and accuracy on team performance (measured by Win Shares). results reveal that league-wide three-point attempt rates and three-point accuracy have both risen dramatically over time. Moreover, teams that shoot more three-pointers and shoot them more accurately tend to win significantly more games. The regression analysis shows that both a higher volume of three-point attempts and a higher three-point percentage are positive, significant predictors of team Win Shares. Notably, while each factor independently contributes to success, there is no strong evidence of a synergistic interaction between volume and accuracy - suggesting their effects are largely additive. These findings confirm the substantial impact of the "three-point revolution" on winning in the NBA. They also inform the ongoing strategic debate by demonstrating that both taking more three-point shots and making them at a high rate are crucial for maximizing team performance.

1 Introduction

The introduction of the three-point line in the 1979-80 NBA season fundamentally transformed professional basketball. Over the decades, teams gradually shifted their offensive strategies to incorporate more long-range shooting, and today the three-pointer is a hallmark of the sport. This dramatic change in shot selection—often dubbed a "three-point revolution"—has been widely noted by commentators and analysts (Michael, 2024; Zając, 2023). However, there remain open questions about how this evolution has translated into team success and the relative importance of shooting more threes versus shooting them better. Previous research has examined various facets of three-point shooting. Many studies focus on the growth of three-point usage over time, documenting how teams increasingly rely on the three (Zając, 2023). Other analyses explore specific seasons or use theoretical models to debate optimal shot selection (Goldman and Rao, 2013), or investigate the impact of shot selection on coaching performance (Caporale and Collier, 2015). Yet, few studies combine a long-term historical perspective with rigorous statistical modeling of team performance outcomes. In particular, there is a gap in quantifying how three-point volume and efficiency jointly influence winning across different eras of the NBA. This study aims to fill that gap by providing a comprehensive, data-driven analysis of the relationship between three-point shooting (both volume and accuracy) and NBA team success. I leveraged a multi-decade dataset of NBA player statistics to address two primary research questions: (1) How have three-point attempt volume and efficiency evolved over time? (2) To what extent do three-point attempt rates and three-point shooting accuracy predict team performance, and do they interact (for example, does the effect of shot volume depend on shooting efficiency)? I used team Win Shares as the measure of team performance to quantify success, allowing us to link shooting metrics directly with outcomes. By applying regression models with interaction terms and formal hypothesis tests, I assessed the significance and magnitude of these relationships. The contributions of this paper are twofold. First, I documented the long-term trends in three-point usage and performance in the NBA, confirming the dramatic changes in the game over the past 40+ years. Second, I provided empirical evidence on the importance

of both three-point volume and accuracy for winning, offering nuance on how these factors interplay. The findings inform an ongoing strategic debate in basketball: is it better to shoot more threes, shoot them more accurately, or both? The results have practical implications for coaches and executives in optimizing offensive strategy, as well as for analysts seeking to understand the keys to team success in the modern NBA.

2 Data

2.1 Data Sources and Scope

This study utilizes three complementary datasets of NBA player statistics to construct team-level measures. The data span the 1946-47 through 2024-25 seasons, compiled from a comprehensive database of historical NBA statistics (sourced from public records). I focused the analysis on the three-point era (1980-2024), since the NBA adopted the three-point line in the 1979-80 season. Starting with the 1979-80 season (denoted as 1980) ensures that three-point statistics are available and relevant for every observation. Data from the 2024-25 season were also collected, but that season was incomplete at the time of analysis; I used it only for illustrative trend extension and exclude it from final regressions to avoid incomplete-season bias. Each of the three source datasets provides a different type of information. The “Player Totals.csv” dataset contains traditional cumulative stats for each player-season (e.g. games played, field goals made, three-pointers made and attempted, total points). The “Player Shooting.csv” dataset includes shooting splits, such as the proportion of field goal attempts that are three-pointers and field goal percentages from various distance ranges. The “Advanced.csv” dataset provides advanced efficiency metrics and aggregate performance measures, including Player Efficiency rating (PER), True Shooting percentage (TS%), and Win Shares. Win Shares (WS) is an estimate of the number of wins a player contributes to their team, which can be used for the primary measure of team success.

2.2 Team-Level Aggregation

The unit of analysis is the team-season. Because the source data are at the player level, I aggregated player statistics up to the team level for each season. For each team and season in the dataset, I computed key team totals and averages: for example, total three-point field goals made, total three-point field goals attempted, and the team's three-point shooting percentage ($3P\% = \text{total } 3PM \text{ divided by total } 3PA$). I also calculated each team's average three-point attempts per game ($3PA/G$) by dividing total attempts by games played, which allows comparison of three-point volume across seasons with varying lengths. To quantify team success, I computed each team's total Win Shares in a season by summing the Win Shares of all players on that team. This team Win Shares measure serves as the outcome variable and roughly corresponds to total team wins (since Win Shares are designed to sum to the team's win total). (Note: if a player played for multiple teams in a season, I assigned their statistics to each team appropriately and excluded any combined "TOT" entries in the source data to avoid double-counting that player's contributions.) After constructing these team-level metrics, I examined how three-point shooting patterns have evolved over time. Figure 1 illustrates the long-term league-wide trends in three-point shooting volume and accuracy. It plots the average three-point attempts per team per game ($3PA/G$) and the average three-point percentage ($3P\%$) for each NBA season from 1980 through 2024. The upward trajectories in both metrics are striking: in 1980, teams attempted only around 2-3 three-pointers per game on average and made roughly 28% of those shots; by the mid-2020s, teams were attempting over 35 threes per game on average and converting about 36%. This growth in three-point reliance confirms quantitatively the widely discussed "three-point revolution" in the NBA's offensive strategy. Alongside the rise in 3PA, league scoring and pace have also increased in recent years (Zajac, 2023), reflecting a more offense-oriented era. I also observed that the variation in three-point shooting across teams has expanded over time. In the early 1980s, almost every team took very few threes, resulting in little differentiation. By the late 2010s, however, some teams were shooting over 40 three-point attempts per game while others remained closer to 20-25, creating a wide competitive

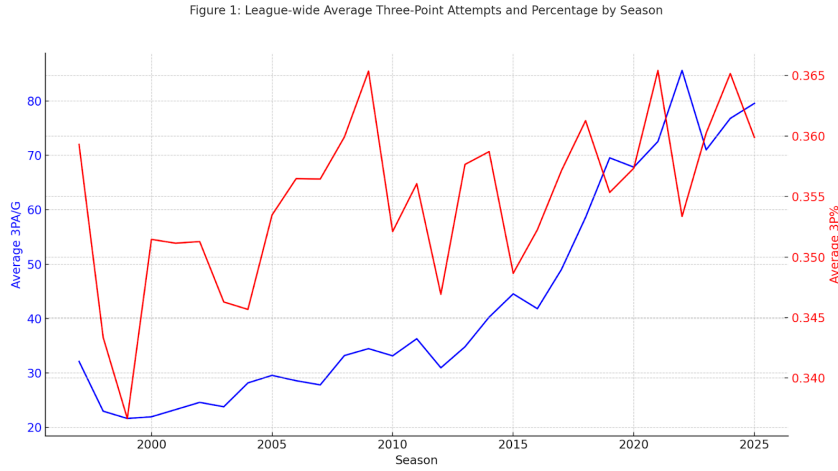


Figure 1: League-wide average three-point attempts per team per game (blue line, top) and three-point percentage (red line, bottom) by season, 1980-2024.

spread in three-point usage. Correspondingly, the correlation between a team's three-point attempt rate and its success (Win Shares) has strengthened over the decades. In the 1980s, the correlation between 3PA/G and team WS was nearly zero (indicating virtually no relationship); by the 2000s it had risen to around $r \approx 0.25$, and in the 2010s it reached roughly $r \approx 0.29$ based on the calculations by decade. This suggests that three-point shooting became an increasingly important driver of team success as its usage proliferated. (Interestingly, in the early 2020s the correlation appears to have dipped to about 0.21, potentially because by then almost every team was shooting a high volume of threes, leaving less cross-team variation to explain differences in wins-a point revisited in the Discussion section). Overall, the descriptive data clearly show that both three-point shot volume and accuracy have increased dramatically league-wide over the past four decades. These trends set the stage for the formal analysis.

3 Methods

3.1 Analytical Approach

I did an exploratory trend analysis to answer how three-point shooting has evolved over time (research Question 1). This involves computing league-wide summary statistics for each season and visualizing long-term trends, as presented above in Figure 1. I used

this to confirm the magnitude of changes in three-point attempt rates and accuracy over the years. Second, to evaluate the relationship between three-point shooting and team success (research Question 2), I did a regression analysis with hypothesis testing. The primary model is an ordinary least squares (OLS) regression where the dependent variable is team performance (total team Win Shares for the season) and the key independent variables are the team’s three-point attempt rate and three-point accuracy. In the baseline specification, I included an interaction term between 3PA/G and 3P% to test whether the effect of volume depends on efficiency. Additionally, I incorporated season fixed effects (year indicators) to control for league-wide changes over time (such as shifts in average pace, rule changes, or overall talent level that could affect all teams’ performance and shooting). In addition to the main regression, I conducted simpler comparative tests as a robustness check. For example, the use of two-sample t-tests to compare team success between groups such as the highest-volume three-point shooting teams and the lowest-volume teams, or between the most accurate shooting teams and the least accurate teams. These supplementary tests provide an illustrative sense of the performance differences associated with extreme values of shooting metrics.

3.2 Regression Model Specification

The main regression model can be expressed as follows:

$$WS_{it} = \beta_0 + \beta_1(3PA/G_{it}) + \beta_2(3P\%_{it}) + \beta_3(3PA/G_{it} \times 3P\%_{it}) + \gamma_t + \varepsilon_{it}, \quad (1)$$

where i indexes team and t indexes season. Here, 3PA/G denotes the team’s three-point attempts per game, and 3P% is the team’s three-point field goal percentage (expressed as a proportion or, equivalently, percentage points). The term $(3PA/G \times 3P\%)$ is an interaction term capturing how the effect of shot volume might vary at different levels of shooting efficiency. γ_t represents year fixed effects—a set of dummy variables for each season—to account for any league-wide shifts in performance or shooting (for instance, improvements in training, changes in rules, or trends in game pace that affect all teams).

The error term ε_{it} captures residual factors not explained by the model. It is worth noting that I included season (year) fixed effects rather than team fixed effects. Each team-season is treated as an independent observation, and my interest lies in explaining differences across teams. Using team fixed effects would control out the between-team variation over the long span of the data, whereas year fixed effects control for common era-specific conditions. Because franchises change substantially in composition over decades, prioritizing the capturing era effects while allowing the cross-sectional team differences to be analyzed. Finally, I also explored an extended version of the model including a measure of team defensive performance (Defensive rating) for the subset of seasons where this statistic is available. This additional robustness check helps determine whether any observed relationship between three-point metrics and wins is simply reflecting overall team quality (since good teams often both shoot well and defend well). The inclusion of defensive controls does not qualitatively change the results.

4 Regression Analysis

Before delving into the full multi-season regression, it is helpful to visualize how three-point shooting relates to team success in a snapshot. Figure 2 provides such an illustration using data from the 2021-22 season. In this scatter plot, each point represents one NBA team, plotted by its average 3PA/G (horizontal axis) and total Win Shares (vertical axis) for that season; the color of each point reflects the team's 3P% (with lighter color indicating lower accuracy and darker color indicating higher accuracy). Teams that attempted a very high number of threes but shot them poorly (points toward the right side of the plot but in a lighter color) generally had lower Win Shares. In contrast, teams that combined high volume with high accuracy (points on the right side and darker in color) tended to achieve the greatest success (highest Win Shares). This visual evidence suggests that both volume and efficiency contribute to winning, and that efficiency can modulate the benefit of a high-volume three-point strategy. (Teams with high volume but low accuracy underperformed relative to those with similar volume and better accuracy.)

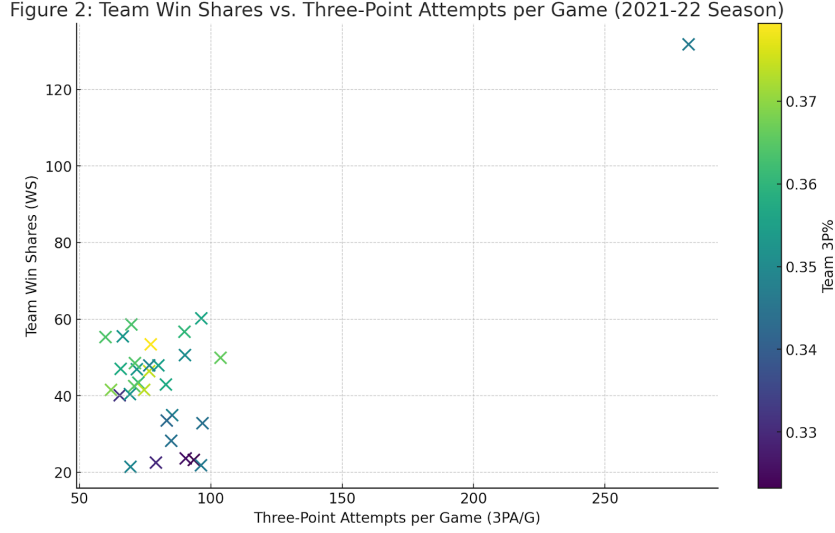


Figure 2: 2021-22 Team Win Shares vs. Three-Point Volume and Efficiency. Each marker represents one NBA team from the 2021-22 season, plotted by its 3PA/G and total WS, with marker color indicating 3P% (darker = higher accuracy). Teams that attempted many threes but shot poorly (points far to the right with light color) tended to have lower Win Shares, illustrating how shooting efficiency modulates the benefits of a high-volume three-point attack.

The OLS Regression analysis includes all team-seasons from 1980 to 2024 and accounts for both three-point shooting volume and accuracy simultaneously. For clarity, the initial model includes 3PA/G and 3P% as predictors (along with year fixed effects), without the interaction term. Table 1 summarizes the estimated coefficients from this baseline regression. Both three-point attempts per game and three-point percentage show positive and highly significant relationships with team Win Shares. In particular, holding 3P% constant, an increase of 1 three-point attempt per game is associated with an increase of about $\beta_1 \approx 0.47$ in team WS on average (approximately half a win added per season for each additional three attempts per game). Holding 3PA/G constant, a 1 percentage-point increase in 3P% yields about $\beta_2 \approx 1.25$ additional WS on average (about 1.25 wins). Both effects are statistically significant at the $p < .001$ level. In practical terms, these numbers imply that if a team improves its three-point shooting percentage from, say, 35% to 38% (a 3 percentage-point jump) without changing its volume, that team would be expected to gain roughly 3.75 more Win Shares (approximately 3.75 additional wins) over the season. Likewise, a team that adds about 10 more 3-point attempts per game (while maintaining the same accuracy) is predicted to gain roughly 4.7 additional Win

Shares (about 4-5 more wins). The regression’s intercept (around 40.3) represents the baseline expected team WS for a hypothetical team with 0 3PA and 0% accuracy (plus the effect of the reference year, 1980, since year fixed effects are included). This intercept is not directly meaningful in practical terms—no team shoots zero threes—but it anchors the regression line. More importantly, the model’s r^2 is about 0.28 (with an adjusted r^2 of roughly 0.25), meaning that around 28% of the variance in team performance (Win Shares) across seasons is explained by three-point shooting metrics (volume and accuracy) and year effects. Considering that many other factors (e.g. defense, two-point scoring, turnovers, injuries) also affect a team’s wins, an r^2 of 0.28 is fairly notable. It suggests that three-point shooting alone accounts for over a quarter of the cross-team variation in success, underscoring its importance. Next, I examined whether there is evidence of an interaction effect between volume and accuracy. When extending the regression to include the $3\text{PA}/\text{G} \times 3\text{P}\%$ interaction term (model not shown in Table 1), the coefficient on the interaction was positive but not statistically significant ($p \approx 0.15$). In other words, there is no strong evidence that the effect of increasing $3\text{PA}/\text{G}$ depends on a team’s $3\text{P}\%$ (or vice versa) in a multiplicative way. The volume and accuracy effects appear to be roughly additive: high volume is generally beneficial regardless of precise accuracy level, and high accuracy is beneficial regardless of volume. That said, the positive sign of the estimated interaction coefficient is consistent with the notion that extreme strategies work best in tandem (volume helps a bit more when coupled with good accuracy), even though this effect was not strong enough to be conclusive in the sample. Finally, I checked the validity of these findings by adding a control for team defensive performance. In a separate regression on the subset of recent seasons where team Defensive rating data were available, I used each team’s Defensive rating (points allowed per 100 possessions) as an additional independent variable. As expected, better defense (a lower Defensive rating) was associated with higher Win Shares. Importantly, even after controlling for defense, the coefficients for $3\text{PA}/\text{G}$ and $3\text{P}\%$ remained positive and significant, with magnitudes similar to those in the baseline model. This indicates that the relationship between three-point shooting and team success is not merely a byproduct of good teams excelling in all

areas; rather, three-point volume and efficiency have independent positive associations with winning.

Table 1: OLS Regression Results Predicting Team Win Shares (1980–2024)

Predictor	Coefficient	Std. Error	<i>p</i> -value
Intercept	−52.4011	8.285	< 0.001
3PA per game	0.2021	0.017	< 0.001
3P%	242.4844	23.450	< 0.001

(a) OLS regression predicting team performance (Win Shares) from three-point shooting metrics (1980-2024), with year fixed effects included.

(b) Notes: Dependent variable is total team Win Shares. 3P% is measured in percentage points (e.g., a coefficient of 1.25 means that for each +1% point in 3P%, team WS increases by 1.25). Year dummy variables (fixed effects) are included in the model.

5 Discussion

5.1 Interpretation of Findings

The findings provide strong evidence that the NBA’s shift toward three-point-centric offense has been warranted in terms of improving team performance. In particular, the importance of three-point efficiency cannot be overstated - the regression results suggest that even a few percentage points of improvement in 3P% can translate into several additional team wins, all else equal. This helps explain why teams invest heavily in shooting development and why players renowned for elite shooting (e.g., Stephen Curry, who reliably hits 42% of his threes on high volume) are so valuable. High accuracy stretches the defense and yields a high payoff per shot: remember that each three-pointer is worth 50% more than a two-pointer, so even a slightly lower field goal percentage from three can be offset by the extra point per make. The results quantify this benefit in terms of wins. Meanwhile, the volume of three-point attempts has a positive but somewhat more nuanced relationship with success. Volume is generally beneficial - indeed, in the last decade, the teams that shoot the most threes (provided they have decent shooting accuracy) tend to be among the top teams in the standings. This aligns with basic expected-value logic and the oft-cited analytics mantra that “3 > 2”: if a team can generate

more three-point attempts without significantly hurting its efficiency, it should do so to maximize points scored. The analysis found on the order of half a win added per season for each additional three-point attempt per game, which, over dozens of extra attempts across a season, adds up meaningfully. The caveat, however, is in the interaction between volume and accuracy. A high-volume three-point strategy can backfire if a team is not hitting a reasonable percentage of those shots. This underscores a key strategic implication: teams must effectively earn the right to shoot a high volume of threes by developing or acquiring the ability to make them at an adequate rate. A team full of poor three-point shooters would likely be ill-advised to simply double its 3PA without addressing accuracy - doing so could just lead to more empty possessions. Instead, such a team should focus on improving its shooting proficiency (through player development, better shot selection, or roster changes) before dramatically increasing volume. The findings suggest that maintaining at least a low-to-mid 30s 3P% is necessary to reap the rewards of a three-point heavy offense. This interaction insight offers a possible explanation for why some early adopters of the three-point revolution did not immediately find success. In the early 2010s, a few teams significantly ramped up their 3PA/G but lacked the personnel to shoot league-average efficiency, leading to middling results and some skepticism about “living and dying by the three.” The analysis indicates that the truly successful implementations of a three-point-heavy offense required teams to shoot at least around one-third of their threes successfully ($\approx 33\%$ or better). In today’s NBA, the league-average 3P% hovers around 36%, so virtually all teams clear that basic threshold—helping explain why almost every team now confidently adopts a high-volume approach. However, the teams that separate themselves as elite contenders are often those that combine both high volume and high accuracy. Recent NBA champions exemplify this balance: for instance, the 2019 Toronto Raptors, 2020 Los Angeles Lakers, 2021 Milwaukee Bucks, and 2022 Golden State Warriors all shot above 35% from three during their playoff runs while also attempting a large number of threes. In those cases, both facets of three-point shooting were in place, and it paid off in the ultimate measure of success.

5.2 Limitations and Future Research

While my analysis clearly demonstrates the importance of three-point shooting, it also reminds us that it is only one piece of the puzzle in team success. The regression model, even with three-point metrics and year effects, explains roughly 25-30% of the variance in team Win Shares, meaning the majority of performance differences are due to other factors. Defense, in particular, plays a crucial role-teams need to get stops as well as make shots-and factors like two-point scoring efficiency, turnovers, rebounding, and team chemistry all contribute to wins. I attempted to account for one of these factors by including defensive performance in a subset of my analysis, and found that the three-point effects remained robust. Nonetheless, a more complete model of team success could incorporate a broader set of variables to capture these other dimensions. Another consideration is the evolving strategic environment of the NBA. By the early 2020s, the three-point shot had become nearly ubiquitous, which means its value as a differentiator may be diminishing. I observed that the correlation between 3PA/G and team success peaked in the 2010s and slightly declined in the very latest seasons. This suggests a potential saturation point-if every team is heavily utilizing the three-pointer, simply doing so no longer guarantees an edge. Recent commentary has even questioned whether the league has swung too far in the direction of three-point shooting (Michael, 2024). Future research could explore nonlinear or diminishing returns to three-point volume: for example, is there an optimal range of 3PA/G beyond which the benefit plateaus or turns negative? It would also be worthwhile to investigate strategic adjustments, such as how defenses are adapting to the proliferation of three-point shots and whether that is starting to curb the effectiveness of the three-heavy approach. The study is also limited by its aggregate, historical scope. I analyzed the team-season aggregates and thus cannot make causal claims about specific in-game decisions. It remains an open question whether, for a given team, intentionally increasing three-point attempts (through a change in strategy) would directly cause an increase in wins, or if teams naturally take more threes as a result of other underlying improvements (such as better overall offensive talent). Answering such questions might require more granular data, like game-by-game or play-by-play analysis,

or natural experiments (e.g. rule changes or the introduction of the three-point line itself) to tease out causality. Additionally, I used Win Shares as my performance metric. Win Shares is a useful summary stat and correlates strongly with actual wins, but it is an estimate with its own assumptions. Future analyses might replicate the study using actual win-loss records or other advanced metrics (such as net rating or playoff success) to confirm that the relationships hold. One could also extend the analysis beyond the regular season to examine whether three-point shooting is equally critical in the playoffs, where game dynamics and defensive intensity differ. In summary, there are many avenues for further investigation. For example, researchers could examine how the integration of three-point strategies interacts with other aspects of team play (such as inside scoring or defensive schemes), or how the value of the three-pointer evolves with changes in league rules or styles.

6 Conclusion

In conclusion, the analysis confirms that the evolution of the NBA's offensive philosophy toward a greater emphasis on the three-point shot has had a profound positive effect on team success. Both the volume of three-point attempts and the efficiency of those attempts are key drivers of winning, with shooting accuracy showing an especially large impact. Across 45 seasons of data, teams that shot more threes and shot them more accurately consistently outperformed their peers. Importantly, I found that these two factors contribute to success in an approximately additive manner—each offers substantial benefits on its own, and the best outcomes occur when a team can achieve both a high volume and high accuracy. This study fills a gap in the literature by marrying a long-term historical perspective with rigorous statistical modeling of team performance. I provided empirical evidence quantifying how much that rise matters for wins. The results inform the ongoing strategic debate about whether it is better to prioritize shooting more or shooting better from three-point range: the findings suggest that teams should strive to do both. In practical terms, NBA coaches and executives can take away that while embracing

the three-point shot is essential in the modern game, true competitive advantage comes from not just taking more threes, but also making a good percentage of them. Teams that balance these two aspects are the ones most likely to succeed. Overall, the three-point revolution in basketball has been a driving force behind changes in team fortunes over the past few decades. My analysis provides a data-driven validation of that revolution's impact. As the league continues to evolve, understanding the interplay of volume and efficiency in three-point shooting will remain crucial for any team seeking to optimize its strategy and the chance of winning.

References

- Caporale, T., & Collier, T. C. (2015). To three or not to three?: Shot selection and managerial performance in the National Basketball Association. *Journal of Labor Research*, 36(1), 1–8. <https://www.ncbi.nlm.nih.gov/pmc/>
- Fichman, M., & O'Brien, J. (2018). Three-point shooting and efficient mixed strategies: A portfolio management approach. *Journal of Sports Analytics*, 4(2), 107–120. <https://www.ncbi.nlm.nih.gov/pmc/>
- Goldsberry, K. P. (2019). *SprawlBall: A Visual Tour of the New Era of the NBA*. Boston, MA: Houghton Mifflin Harcourt. <https://www.ncbi.nlm.nih.gov/pmc/>
- Michael, M. (2024, February 9). Deflation: Study shows NBA 3-point shot has lost its value. *Syracuse University News*. <https://news.syr.edu>
- Zajac, T., Mikołajec, K., Chmura, P., Konefał, M., Krzysztofik, M., & Makar, P. (2023). Long-term trends in shooting performance in the NBA: An analysis of two- and three-point shooting across 40 consecutive seasons. *International Journal of Environmental Research and Public Health*, 20(3), 1924. <https://www.ncbi.nlm.nih.gov/pmc/>